



FACULTY OF ARCHITECTURE & TOWN PLANNING

Aror University

Fall 2026

Course Title:	Digital Logic Design (DLD)
Course Code:	CSC214
Credit Hours:	(2+1)
Course Instructor:	Muzaffar Hussain
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Description:

Digital Logic Design (DLD) is a branch of electronics that deals with the study and design of digital circuits—systems that use signals having only two states: 0 (Low / OFF), 1 (High / ON).

It provides the foundation of modern computers, microprocessors, and digital systems. In this subject, we learn how basic electronic components, like logic gates, can be combined to perform logical decisions, arithmetic operations, data storage, and communication tasks.

Aim and Objectives

Understand the difference between analog and digital systems, binary digits, and logic levels. Apply number systems and conversions (binary, octal, decimal, hexadecimal) in digital circuit design.

Analyze basic logic gates, Boolean algebra, and simplification techniques.

Design combinational logic circuits such as adders, subtractors, multiplexers, demultiplexers, encoders, and decoders.

Construct sequential circuits such as latches, flip-flops, counters, and registers.

Implement digital circuits using Karnaugh maps (K-maps) and other minimization techniques. Relate the use of digital logic to real-life applications in computer systems, communication, and AI.

Develop problem-solving skills for troubleshooting and debugging digital systems.

Utilize simulation tools and laboratory experiments to validate theoretical concepts in a practical setting.



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Assessment Criteria:

S. No	Assessment Activities	Percentage	Total Activities	CLO#
1.	Quizzes/ Assignments (2 Quizzes & 2 Assignments)	30%	4	1,2
2.	Mid Term Exam	30%	1	1
3.	Final Exam	40%	1	2

CLOs:

After completing the “Applied Physics” Course, each student will be able to:

CLO	Description	Taxonomy Level	PLO
1	UNDERSTAND concepts related to basic Digital Logic Design (DLD).	C2	1
2	APPLY fundament concepts of Digital logic design.	C3	2



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Week	LEC#	Sub-Topic	CLO	Domain	Reference	% Course Covered
Week-1	LEC-1	- Introduction to the Class & Subject - Introductory Analog and Digital Concepts - Binary digits	CLO-01	C2	Floyd Thomas Prentice Hall, USA.	3%
	LEC-2	- Logic levels - Digital waveform (periodic and non-periodic) - Basic Logic Operations.	CLO-01	C2	Floyd Thomas Prentice Hall, USA.	6%
Week-2	LEC-3	Number Systems and conversions (Binary, Decimal, Hexadecimal, Octal)	CLO-01	C2	Floyd Thomas Prentice Hall, USA.	9%
	LEC-4	- Signed Numbers, - Arithmetic Operations with Signed Numbers	CLO-01	C2	Floyd Thomas Prentice Hall, USA.	12%
Week-3	LEC-5	- Binary Arithmetic, - Digital Codes (Gray Code, Alphanumeric Codes, Unicode) - Error Codes (Parity Method for Error Detection and Cyclic Redundancy Check)	CLO-01	C2	Floyd Thomas Prentice Hall, USA.	16%
	LEC-6	- Logic Gates - Truth Table	CLO-01	C2	Floyd Thomas Prentice Hall, USA.	18%
Week-4	LEC-7	- Boolean Algebra, - Apply the basic laws and rules of Boolean Algebra to simplify Boolean expressions - DE Morgan's theorems.	CLO-01	C2	Floyd Thomas Prentice Hall, USA.	22%
	LEC-8	- Sum-of-Product (SOP) and the Product-of Sum (POS) expressions - converting a Boolean expression to a truth table and vice versa.	CLO-01	C2	Floyd Thomas Prentice Hall, USA.	25%
	Quizz-01 & Assignment 01					
Week-5	LEC-9	- The Karnaugh Map - simplification using Karnaugh Map - The Karnaugh Map Examples	CLO-01	C2	Floyd Thomas Prentice Hall, USA.	28%



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	LEC-10	Quine McCluskey Method	CLO-01	C2	Floyd Thomas Prentice Hall,USA.	31%
Week-6	LEC-11	Combinational Logic Analysis	CLO-01	C2	Floyd Thomas Prentice Hall,USA.	34%
	LEC-12	Basic Combinational Logic Circuits	CLO-01	C2	Floyd Thomas Prentice Hall,USA.	38%
Week-7	LEC-13	Implementing Combinational Logic	CLO-01	C2	Floyd Thomas Prentice Hall,USA.	41%
	LEC-14	The Universal Property of NAND and NOR Gates	CLO-01	C2	Floyd Thomas Prentice Hall,USA.	44%
Week-8	LEC-15	Combinational Logic Using NAND and NOR Gates	CLO-01	C2	Floyd Thomas Prentice Hall,USA.	47%
	LEC-16	Pulse Waveform Operation.	CLO-01	C2	Floyd Thomas Prentice Hall,USA.	50%
	Quizz-02 & Assignment 02					
Mid Term Examination						
Week-9	LEC-17	- Functions of Combinational Logic - Half and Full Adders - Parallel Binary Adders	CLO-02	C3	Compton Floyd Thomas Prentice Hall,USA.	53
	LEC-18	- Ripple Carry and Look-Ahead Carry Adders - Comparators	CLO-02	C3	Floyd Thomas Prentice Hall,USA.	56
Week-10	LEC-19	- Decoders - Encoders - Code Converters	CLO-02	C3	Floyd Thomas Prentice Hall,USA.	59
	LEC-20	- Multiplexers (Data Selectors) - Demultiplexers - Parity Generators/Checkers.	CLO-02	C3	Floyd Thomas Prentice Hall,USA.	63
Week-11	LEC-21	- Latches, Flip-Flops, and Timers - Latches: Introduction - S-R Latch	CLO-02	C3	Floyd Thomas Prentice Hall,USA.	66
	LEC-22	- Gated S-R Latch - Gated D-Latch.	CLO-02	C3	Floyd Thomas Prentice Hall,USA.	69
Week-12	LEC-23	- Flip Flops: - S-R Flip Flop - J-K Flip Flop	CLO-02	C3	Floyd Thomas Prentice Hall, USA.	72



	LEC-24	• D-Flip Flop • T-Flip Flop • Applications	CLO-02	C3	Floyd Thomas Prentice Hall,USA.	75
	Quizz-03 & Assignment 03					
Week-13	LEC-25	• Shift Register Operations • Types of Shift Register Data I/Os	CLO-02	C3	Floyd Thomas Prentice Hall,USA.	78
	LEC-26	• Bidirectional Shift Registers • Shift Register Counters • Shift Register Applications.	CLO-02	C3	Floyd Thomas Prentice Hall,USA.	81
Week-14	LEC-27	• Counters	CLO-02	C3	Floyd Thomas Prentice Hall,USA.	84
	LEC-28	• Finite State Machines • Asynchronous Counters	CLO-02	C3	Floyd Thomas Prentice Hall,USA.	88
Week-15	LEC-29	• Synchronous Counters • Up/Down Synchronous Counters	CLO-02	C3	Floyd Thomas Prentice Hall,USA.	91
	LEC-30	• Cascaded Counters • Counter Applications	CLO-02	C3	Floyd Thomas Prentice Hall,USA.	94
Week-16	LEC-31	Final Review and Exam Preparation	CLO-02	C3	Floyd Thomas Prentice Hall,USA.	97
	LEC-32	Course Wrap-up and Final Discussion	CLO-02	C3	Floyd Thomas Prentice Hall,USA.	100
	Quizz-04 & Assignment 04					
Final Examination						